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Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



**TUTORLY: AN INTELLEAGENT TUTOR MATCHING AND
ACADEMIC PROGRESS MONITORING SYSTEMS
FOR LEARNERS IN NASUGBU, BATANGAS**

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

INTRODUCTION

Technology has consistently evolved and has significantly altered the ways in which individuals communicate, work, and study. Digital technologies have fundamentally transformed the character and scope of education, prompting educational systems globally to implement strategies and policies for digital integration [1]. Advancements in digital technologies have allowed businesses and institutions to enhance efficiency, accessibility, and the quality of services offered to users. The successful integration of information and communication technology resources across various educational levels allows stakeholders to develop interactive learning environments that enhance student engagement and elevate learning outcomes [2]. The use of technological solutions in the education industry has revolutionized traditional learning environments into more adaptable and information-driven systems that benefit both students and educators. The education technology industry has focused on developing and releasing digital learning platforms, learning management systems, and customized tutoring settings that enable students to obtain educational support beyond limitations imposed by traditional classrooms [3].

Artificial intelligence has greatly contributed to the advancement of current educational technologies. A key advancement in this field is the rise of Intelligent Tutoring Systems (ITS), which replicate the assistance offered by human tutors through computational models and artificial intelligence methodologies. AI-driven Intelligent Tutoring Systems offer significant potential to revolutionize education by facilitating personalized instruction and adaptive feedback to assist learners [4]. These systems evaluate student behavior and performance data to deliver instructional support that enhances the learning experience. Adaptive learning systems are designed to modify educational content and instructional techniques based on learner performance and behavior. Artificial intelligence and machine learning algorithms utilized in adaptive learning systems have demonstrated the ability to optimize learning paths, enhance student engagement, and elevate academic performance [5]. Moreover, learning analytics has become a crucial instrument for extracting insights from student learning practices. Educators can use learning analytics in order to address student engagement issues related to online learning, therefore enhancing learning quality and performance [6]. Educational recommender systems have been extensively utilized in online learning platforms to deliver individualized recommendations that match learners' academic standards and preferences [7].

Despite these technological advancements, students worldwide continue to face significant difficulties in accessing quality tutoring services that meet their academic needs. Tutoring programs yield consistent and substantial positive impacts on learning outcomes; however, access to these programs remains unequal across different populations and educational contexts [8]. Barriers such as limited tutor availability, inadequate coordination systems, and poor communication between families and service providers continue to restrict many students from receiving the academic support they require [8]. Globally, the absence of structured digital platforms for managing tutoring services has resulted in inefficient systems that fail to address the diverse learning needs of students across different educational contexts.

Studies in regions like Europe and Asia have emphasized the increasing demand for personalized tutoring services, especially in competitive academic settings. Research indicates that shadow education — the informal practice of private extra tutoring — has consistently expanded over the past decade and occurs frequently across various educational environments, especially in East Asia [9]. The participation in private supplementary tutoring has risen across Asia; despite this, these systems mainly lack regulation, digitization, and systematic means for monitoring student learning progress [9]. The lack of digital infrastructure for

monitoring the results of private tutoring makes it challenging for parents and educators in Europe and Asia to assess the effectiveness of tutoring sessions in achieving significant learning outcomes. These gaps emphasize the necessity for intelligent technologies that can facilitate the alignment of tutors with students and the systematic assessment of academic advancement.

In the Philippines, the challenge of accessing structured and quality tutoring services is particularly evident. The country faces a shortage of approximately 86,000 teachers, with a current student-to-teacher ratio of 1:40, far exceeding the ideal ratio of 1:30 recommended by the Department of Education [11]. This shortage places additional pressure on students to seek external academic support, yet the tutoring landscape in the Philippines remains largely informal and lacks digital infrastructure for systematic coordination and monitoring. The Philippines ranked 77th out of 81 countries in the 2022 Programme for International Student Assessment (PISA), scoring significantly below the global average in mathematics, reading, and science, with the Department of Education acknowledging a five- to six-year lag in learning competencies [12]. These learning challenges further underscore the urgent need for accessible and structured tutoring services supported by digital platforms that can facilitate proper coordination, matching, and progress monitoring.

This issue is most evident at the local level, where the lack of a centralized digital platform for tutoring services hinders effective coordination among students, tutors, and parents. Studies on educational recommender systems and tutoring platforms have observed that even technologically advanced platforms face challenges in establishing consistent and systematic mechanisms for monitoring academic progress [7]. Numerous local tutoring arrangements lack adequate monitoring tools, preventing parents and educators from assessing the effectiveness of tutoring on improving student achievement. The lack of systematic progress reporting, compatibility-driven matching systems, and centralized coordination tools continues in limiting the total effectiveness and accessibility of local tutoring services.

In response to these challenges, this study proposes the development of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System. The proposed system is to establish a centralized web-based platform that connects students, tutors, and parents, while enabling intelligent matching of tutors and students based on compatibility attributes like subject proficiency, availability, learning difficulty, tutor ratings, and budget alignment. The system will incorporate capabilities enabling instructors to deliver quizzes and assessments, as well as generate progress reports to track students' academic development. The suggested system aims to enhance the efficiency of tutoring services by integrating

intelligent tutoring systems, recommendation algorithms, and learning analytics, while offering a systematic method for monitoring academic progress. Utilizing modern educational technologies, the system seeks to enhance communication among stakeholders and increase the overall tutoring experience for students.

Background of the Study

Global education systems persistently encounter difficulties in ensuring that pupils attain the expected criteria for academic proficiency. A major worry in recent years is the expanding educational disparity among pupils, especially following the disruptions induced by the COVID-19 pandemic. The closure of schools and transition to remote learning adversely impacted pupils' capacity to maintain academic progress, leading to quantifiable learning deficits. Research reveals that students experienced a 0.8 standard deviation reduction in educational achievement test scores during the epidemic, reflecting a significant loss in academic performance [13]. The learning deficits have rendered it essential to implement supplementary academic support systems to assist pupils in recuperating lost knowledge.

Tutoring has emerged as a popular approach for addressing learning deficiencies and offering tailored academic support. Unlike traditional classroom

settings where teachers must attend to many students simultaneously, tutoring allows for more individualized instruction tailored to the learner's needs. Research characterizes tutoring as a variant of shadow education that complements conventional schooling by aiding students in enhancing their comprehension of academic subjects and preparing for assessments [14].

International studies provide evidence of the increasing demand for tutoring services. In a research investigating private tutoring practices, 357 of 503 students (71%) indicated they received private tutoring to improve their academic performance and prepare for competitive educational opportunities [14]. The research revealed that 29% of parents allocated between 31,000 and 40,000 tenge (about US\$67–87) monthly for tutoring services, demonstrating the considerable financial commitment families are prepared to make to enhance their children's education [14]. Nonetheless, in spite of its advantages, the study also uncovered difficulties inside the tutoring environment. For example, 48% of students indicated a lack of familiarity with their allocated tutors, underscoring concerns over tutor matching, transparency, and quality assurance [14].

Given the global learning crisis, numerous educational interventions have been implemented to assist students in regaining the knowledge they have lost. Empirical research on online tutoring services has demonstrated that tutoring interventions can markedly enhance academic achievement. Findings demonstrate

that tutoring programs enhanced exam performance by 0.4 standard deviations, effectively mitigating approximately 50% of the learning deficit resulting from school closures [13]. These findings underscore the efficacy of tutoring as a method for mitigating learning deficiencies and enhancing student performance.

Tutoring support is also evident in the Philippine educational context, as is the issue of learning differences. Reports suggest that some Filipino students resumed in-person classrooms unable to recollect or comprehend teachings from the preceding academic year, attributed to the constraints of distant learning during the epidemic [15]. National education data reveals a significant decline in student competency as they progress through their schooling, with proficiency rates plummeting from roughly 30% in Grade 3 to a mere 0.47% by Grade 12, thereby illustrating the extent of educational inequities within the country's system [16].

To address these challenges, the Philippines has witnessed the implementation of various initiatives designed to support students facing academic difficulties. Both governmental and community organizations have instituted tutoring and literacy programs, with the objective of improving fundamental skills and mitigating learning deficits. These initiatives emphasize the crucial role of tutoring in fostering literacy, academic success, and overall educational outcomes. Despite these efforts, however, access to reliable tutoring services remains inconsistent for a considerable number of students. Students frequently rely on informal networks,

personal recommendations, or tutoring centers to secure academic assistance, a practice that does not always guarantee consistent accessibility or quality. Moreover, many tutoring arrangements lack robust mechanisms for effectively monitoring student academic progress or evaluating the impact of tutoring sessions.

An additional hurdle stems from the fragmented and unstructured nature of tutoring services, thereby hindering students' ability to connect with suitable tutors possessing the requisite expertise for their academic needs. Absent structured systems for tutor matching and progress assessment, students may struggle to identify tutors specializing in their specific subject areas. Similarly, tutors may face challenges in finding students in need of their services, potentially resulting in inefficiencies within the tutoring ecosystem. Consequently, there is a demonstrable need to develop more organized and technology-based solutions designed to improve the accessibility and efficacy of tutoring services, given these existing challenges.

There is a clear demand to create more organized and technology-driven solutions that can enhance the accessibility and effectiveness of tutoring services in light of these challenges. Streamlining the process of connecting students with appropriate educators and monitoring academic progress over time is a potential benefit of intelligent systems and digital platforms. With the integration of

academic progress monitoring features and intelligent tutor recommendation mechanisms, educators can more effectively monitor learning outcomes, while students can receive more personalized learning support.

Thus, this investigation suggests the implementation of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for Learners in Nasugbu, Batangas. The system aims to fix deficiencies in tutoring accessibility and efficacy by offering a systematic platform for optimal tutor-student pairing and academic progress evaluation, hence enhancing student learning results. The learning deficits have rendered it essential to implement supplementary academic support systems to assist pupils in recuperating lost knowledge.

Tutoring has emerged as a popular approach for addressing learning deficiencies and offering tailored academic support. Unlike traditional classroom settings where teachers must attend to many students simultaneously, tutoring allows for more individualized instruction tailored to the learner's needs. Research characterizes tutoring as a variant of shadow education that complements conventional schooling by aiding students in enhancing their comprehension of academic subjects and preparing for assessments [14].

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Given the global learning crisis, numerous educational interventions have been implemented to assist students in regaining the knowledge they have lost. Empirical research on online tutoring services has demonstrated that tutoring interventions can markedly enhance academic achievement. Findings demonstrate that tutoring programs enhanced exam performance by 0.4 standard deviations, effectively mitigating approximately 50% of the learning deficit resulting from school closures [13]. These findings underscore the efficacy of tutoring as a method for mitigating learning deficiencies and enhancing student performance.

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preceding academic year, attributed to the constraints of distant learning during the epidemic [15]. National education statistics indicate that student competency rates markedly decrease over grade levels, falling from approximately 30% in Grade 3 to merely 0.47% by Grade 12, highlighting the severity of learning disparities within the nation's education system [16].

In response to these issues, numerous efforts have been implemented in the Philippines to assist struggling learners. Government and community groups have established tutoring and literacy efforts aimed at enhancing kids' core skills and addressing learning gaps. These programs underscore the significance of tutoring in enhancing literacy, academic achievement, and overall educational results. Notwithstanding these endeavors, access to dependable tutoring services continues to be irregular for numerous learners. Students sometimes depend on informal networks, personal referrals, or tutoring centers to obtain academic support, which may not consistently ensure accessibility or quality. Furthermore, numerous tutoring arrangements lack systems for adequately tracking student academic progress or assessing the efficacy of tutoring sessions.

An additional challenge is the fragmentation and lack of structure of tutoring services, which complicates the process of students establishing connections with qualified tutors who are relevant to their academic requirements. Without structured systems for tutor matching and progress monitoring, students may

encounter difficulty in locating tutors who specialize in the subjects they require assistance with. Likewise, tutors may encounter difficulties in locating pupils who require their services, which can lead to inefficiencies in the tutoring ecosystem.

There is a clear demand to create more organized and technology-driven solutions that can enhance the accessibility and effectiveness of tutoring services in light of these challenges. Intelligent systems and digital platforms offer the potential to streamline the process of connecting students with suitable educators and tracking academic progress. By incorporating features for monitoring academic advancement and intelligent tutor recommendation systems, educators can more effectively assess learning outcomes, and students can receive more tailored learning support. Consequently, this study proposes the implementation of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for Learners in Nasugbu, Batangas. The system is designed to address shortcomings in tutoring accessibility and effectiveness by providing a structured platform for optimal tutor-student pairing and academic progress evaluation, thereby improving student learning outcomes.

Consequently, this study advocates for the deployment of Tutorly, an Intelligent Tutor Matching and Academic Progress Monitoring System, specifically designed for learners in Nasugbu, Batangas. The primary objective of this system is to address existing shortcomings in tutoring accessibility and

effectiveness. It achieves this by providing a structured platform that facilitates optimal tutor-student pairings and comprehensive academic progress assessments, thereby fostering improved student learning outcomes.

OBJECTIVES OF THE STUDY

The primary objective of this study is to design and develop Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for learners in Nasugbu, Batangas that utilizes learning difficulty profiling and a rule-based compatibility algorithm to improve tutor–student matching, support academic progress monitoring, and provide an organized platform for communication between students, tutors, and parents.

To achieve this general goal, the study sets the following specific objectives:

1. To develop a centralized platform that connects students, parents, and tutors in Nasugbu, Batangas to improve access to tutoring services.
2. To design a learning difficulty profiling module that identifies students' academic challenges using diagnostic questions, quizzes, tutor assessments, and academic records.

3. To implement an intelligent tutor recommendation system that suggests compatible tutors based on subject expertise, availability, location, learning difficulty, ratings, and budget compatibility.
4. To develop a student progress monitoring system that records tutoring sessions, quiz and examination scores, tutor evaluations, attendance, and visual progress reports.
5. To provide structured communication and assessment tools, including messaging features and an online quiz and examination module to support tutoring sessions.
6. To implement a tutor verification, rating, and secure payment system to ensure platform credibility and reliable transaction management.
7. To design an administrative dashboard that allows administrators to manage accounts, monitor system activities, and generate analytical reports.

SIGNIFICANCE OF THE STUDY

The implementation of “Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System” is designed to improve the accessibility and organization of tutoring services through a centralized digital platform. The system aims to facilitate efficient tutor-student matching, monitor academic progress, and

provide structured communication among students, tutors, and parents. The proposed system is expected to benefit several stakeholders, including students, tutors, parents, educational institutions, and future researchers.

Batangas State University. The university will benefit from this study as it demonstrates the capacity of its students and faculty to develop intelligent and technology-driven solutions that address real-world educational challenges. Tutorly contributes to the university's commitment to research, innovation, and academic excellence by producing a system that promotes the use of advanced technologies in education. The study also reflects the university's goal of producing graduates who are capable of developing meaningful technological solutions that serve the needs of the academic community.

College of Informatics and Computing Sciences (CICS). The college will directly benefit from this study as Tutorly serves as a demonstration of the practical application of computing and information systems knowledge in solving educational problems. The system showcases the competencies of CICS students in system development, artificial intelligence, and data-driven solutions. Furthermore, the study may serve as a reference model for future capstone and thesis projects within the college that involve the development of intelligent educational platforms.

General Public. Students, tutors, and parents who are part of the broader academic community stand to benefit from the deployment of Tutorly. Students will gain easier access to qualified tutors through an intelligent matching system that considers subject expertise, availability, and learning needs, while also being able to monitor their own academic progress through performance reports and assessment results. Tutors will have a structured platform to manage sessions, track student development, and expand their reach to learners who require their expertise. Parents and guardians will be able to stay informed about their child's academic progress through transparent access to tutoring records, evaluation results, and progress reports, enabling them to better support their child's learning journey.

Future Researchers. This study may serve as a valuable reference for future researchers who are interested in developing intelligent tutoring platforms, recommendation systems, or learning analytics technologies in the field of education. The findings, methodology, and system design of Tutorly may contribute to further studies related to personalized learning systems, academic progress monitoring platforms, and intelligent educational technologies. Future researchers may also build upon the limitations of this study to develop more advanced and comprehensive solutions for tutoring service management.

SCOPE AND LIMITATION OF THE STUDY

This study focuses on the development of Tutorly: An Intelligent Tutor Matching and Academic Progress Monitoring System for Learners in Nasugbu, Batangas. The system is designed as a web-based platform that connects students, tutors, and parents in one centralized environment that supports organized tutoring services. The main goal of the system is to help students find suitable tutors based on their academic needs while providing tools that allow tutors and parents to monitor student learning progress.

The system includes several modules that support the tutoring process. One of the main modules is the Learning Difficulty Profiling module, which allows students to answer structured questions that help identify common learning challenges such as reading comprehension, mathematical problem solving, study habits, and focus during lessons. The responses gathered from this module are used by the Intelligent Tutor Recommendation module, which suggests tutors based on compatibility factors such as subject specialization, tutor experience, availability schedule, location, tutor ratings, and tutoring rates. The system provides a list of recommended tutors together with compatibility scores to help students and parents select suitable tutors.

The system also includes a Student Progress Monitoring module that allows tutors to record tutoring session details, topics discussed, and student performance after each session. These records are stored in the system and presented through progress summaries so that students and parents can monitor learning improvement over time.

Another important feature is the Booking and Scheduling module, which allows students to view tutor availability and book tutoring sessions through a calendar-based interface. The system also provides a Messaging and Communication module that enables tutors and students to communicate within the platform while keeping records of their conversations.

To maintain reliability within the system, a Tutor Verification and Rating module is included. Tutors are required to upload their credentials for administrator verification before they are allowed to provide tutoring services. After tutoring sessions, students and parents may submit ratings and feedback about the tutor's performance.

The system also includes an Online Quiz and Examination module, which allows tutors to create quizzes and assessments to evaluate student understanding of lessons. In addition, the platform provides a Payment Management module,

which displays tutoring session costs and records the payment status for each scheduled session.

The Tutorly system is developed as a web-based application that can be accessed through modern web browsers using desktop computers, laptops, or mobile devices with internet access.

Despite the expected benefits of the system, the study has several limitations. The system is designed specifically for basic education students within the Municipality of Nasugbu, Batangas, and it may require further development before it can be implemented in other locations or educational levels. The tutor recommendation feature uses a rule-based compatibility algorithm, meaning that tutor suggestions are based on predefined criteria rather than advanced machine learning techniques. The Learning Difficulty Profiling module uses structured questionnaires to identify possible learning challenges, and it does not serve as a professional educational or psychological diagnostic tool. Additionally, the payment management feature only records and tracks payment transactions within the system and does not fully integrate with external financial institutions or payment gateways for real-time financial processing. Finally, the study focuses on the development and functionality of the platform, and the long-term academic improvement of students using the tutoring services is beyond the scope of this research.

DEFINITION OF TERMS

To avoid ambiguity, here are the following terms used in this paper:

Academic Progress Monitoring. This refers to the systematic process of tracking and evaluating a student's academic performance over time. In this study, it refers to the Tutorly system's ability to record tutoring sessions, quiz results, tutor evaluations, and progress reports to monitor students' learning improvement.

Adaptive Learning System. This refers to a technology-supported learning approach that adjusts instructional content and learning support based on student performance and behavior. In this study, it refers to the Tutorly system's capability to provide personalized learning support through tutor recommendations and performance monitoring.

Administrator Dashboard. This refers to the system interface used by administrators to manage and monitor system operations. In this study, it refers to the module in Tutorly where administrators verify tutor credentials, manage user accounts, monitor system activity, and generate reports.

Artificial Intelligence (AI). This refers to computer-based technologies that simulate human intelligence to perform tasks such as analysis, decision-making,

and pattern recognition. In this study, it refers to the intelligent mechanisms used in Tutorly to assist in tutor recommendation and data analysis.

Booking and Scheduling Module. This refers to the system component that allows users to arrange and manage appointments or service schedules. In this study, it refers to the feature in Tutorly where students can view tutor availability and book tutoring sessions using a calendar-based interface.

Compatibility Algorithm. This refers to a computational method used to evaluate relationships or matches between different sets of data. In this study, it refers to the rule-based process in Tutorly that calculates the suitability between students and tutors based on subject expertise, availability, ratings, learning difficulty, and budget.

Intelligent Tutor Matching. This refers to a system process that automatically recommends appropriate tutors to students based on specific compatibility factors. In this study, it refers to Tutorly's capability to analyze student learning needs and match them with suitable tutors using predefined criteria.

Intelligent Tutoring System (ITS). This refers to a computer-based educational system that provides personalized learning support similar to a human tutor. In this study, it refers to the intelligent features integrated in Tutorly that assist in guiding students through tutoring support and academic monitoring.

Learning Analytics. This refers to the collection, measurement, and analysis of data related to learning activities. In this study, it refers to Tutorly's ability to analyze student performance data, quiz scores, and tutoring records to evaluate academic progress.

Learning Difficulty Profiling. This refers to the process of identifying students' learning challenges based on their responses and performance data. In this study, it refers to the Tutorly module that gathers diagnostic information from students to determine their academic difficulties and recommend appropriate tutors.

Messaging and Communication Module. This refers to the system feature that allows users to exchange messages and communicate within the platform. In this study, it refers to the Tutorly feature that enables students and tutors to coordinate tutoring sessions and discuss academic concerns through internal messaging.

Online Quiz and Examination Module. This refers to a digital assessment tool used to evaluate students' knowledge and learning progress. In this study, it refers to the Tutorly feature that allows tutors to create quizzes and examinations to measure students' understanding of lessons.

Online Tutoring Platform. This refers to a digital system that allows students and tutors to connect and conduct tutoring services through the internet. In this

study, it refers to the Tutorly web-based platform that facilitates tutor matching, communication, session scheduling, and progress monitoring.

Payment Management System. This refers to a digital mechanism used to track and record financial transactions related to services. In this study, it refers to the Tutorly feature that records tutoring session fees and tracks the payment status of scheduled tutoring sessions.

Personalized Learning. This refers to an educational approach where learning support and instruction are tailored to the individual needs and abilities of each learner. In this study, it refers to the Tutorly system's ability to recommend tutors and monitor progress based on a student's specific academic challenges.

Progress Report. This refers to a structured summary of a learner's academic performance over a period of time. In this study, it refers to the system-generated report in Tutorly that displays quiz scores, tutoring session records, and overall learning improvement.

Recommendation System. This refers to a computational system designed to suggest relevant options based on user data and preferences. In this study, it refers to the tutor recommendation mechanism in Tutorly that suggests compatible tutors to students.

Rule-Based System. This refers to a decision-making system that operates using predefined logical rules and conditions. In this study, it refers to the matching logic used in Tutorly to determine tutor compatibility based on specific criteria.

Tutor Rating System. This refers to a feedback mechanism used to evaluate service quality based on user experiences. In this study, it refers to the Tutorly feature that allows students and parents to rate tutors based on teaching effectiveness and reliability after tutoring sessions.

Tutor Verification System. This refers to a validation process used to confirm the authenticity and qualifications of service providers. In this study, it refers to the Tutorly feature where tutors must submit credentials for administrator approval before offering tutoring services.

Tutorly System. This refers to the proposed intelligent tutor matching and academic progress monitoring platform developed in this study to connect students, tutors, and parents while supporting organized tutoring services and learning analytics.

Web-Based Application. This refers to a software application that operates on a web server and is accessed through internet browsers. In this study, it refers to the

Tutorly platform that can be accessed through computers, laptops, or mobile devices with an internet connection.